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ABSTRACT

1. INTRODUCTION

1.1 PROJECT INTRODUCTION

2. METHODOLOGY

AGILE MODEL

Agile SDLC model is a combination of iterative and incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software product. Agile Methods break the product into small incremental builds. These builds are provided in iterations. Each iteration typically lasts from about one to three weeks. At the end of the iteration, a working product is displayed to the customer and important stakeholders. Every iteration involves cross functional teams working simultaneously on various areas like.

- Planning
- Requirements Analysis
- Design
- Coding
- Unit Testing and
- Acceptance Testing.

What is Agile?

Agile model believes that every project needs to be handled differently and the existing methods need to be tailored to best suit the project requirements. In Agile, the tasks are divided to time boxes (small time frames) to deliver specific features for a release. Iterative approach is taken and working software build is delivered after each iteration. Each build is incremental in terms of features; the final build holds all the features required by the customer.

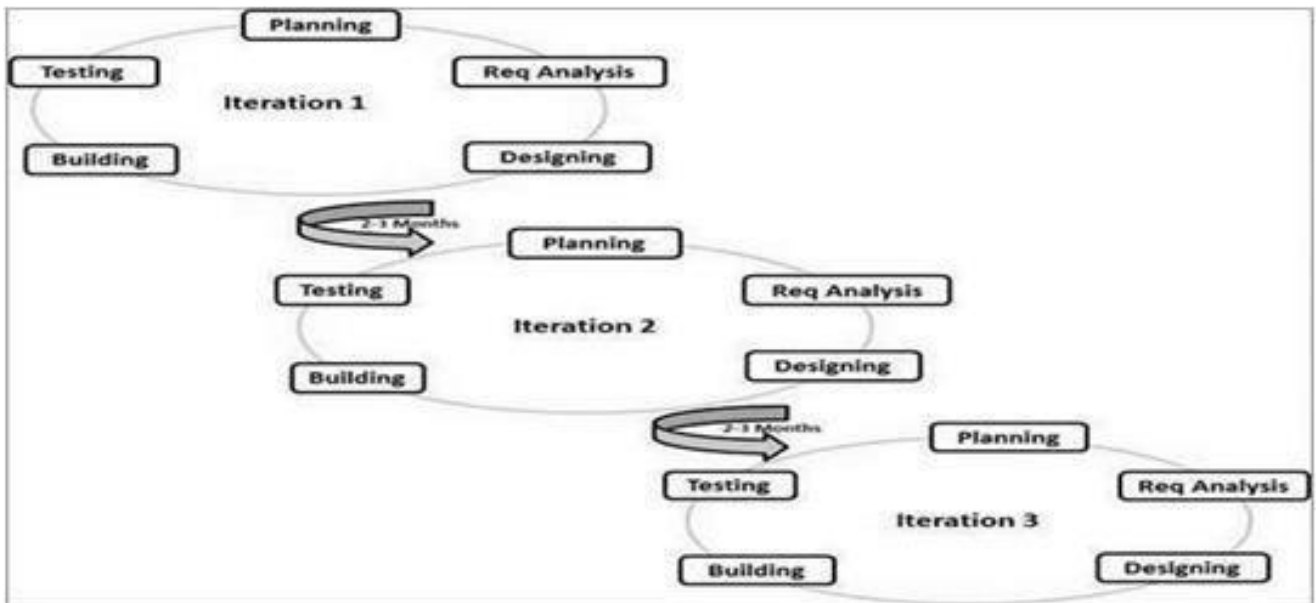


FIG.1 Graphical illustration of the Agile Model

The Agile thought process had started early in the software development and started becoming popular with time due to its flexibility and adaptability. The most popular Agile methods include Rational Unified Process(1994),Scrum (1995), Crystal Clear, Extreme Programming(1996),Adaptive Software Development, Feature Driven Development, and Dynamic Systems Development Method (DSDM) (1995).These are now collectively referred to as Agile Methodologies, after the Agile Manifesto was published in 2001.

Following are the Agile Manifesto principles –

Individuals and interactions – In Agile development, self-organization and motivation are important, as are interactions like co-location and pair programming.

Working software – Demo working software is considered the best means of communication with the customers to understand their requirements, instead of just depending on documentation.

Customer collaboration – As the requirements cannot be gathered completely in the beginning of the project due to various factors, continuous customer interaction is very important to get proper product requirements.

Responding to change – Agile Development is focused on quick responses to change and continuous development.

3. SYSTEM ANALYSIS

3.1 SYSTEM ANALYSIS

System Analysis is concerned with analyzing, designing, implementing and evaluating information system in our organization .It is carried out to make the system more effective either by modification or by substantial redesign. In system analysis we identify the problem, study the alternative solution and select the most suitable solution, which meet the technical economic and social demands for analysis, various tools such as dataflow diagram, interviews on site observation, questionnaires etc, are used. System analysis process is also called a life cycle methodology since it relates to four significant phases in life cycle of all information system. They are

1. System Analysis / Study Phase.
2. System Design / Design phase.
3. System Development / Development Phase.
4. Testing and implementation / Operation Phase.

All activities associated with each life cycle phase must be performed, managed and documented. So system analysis is the performance, management and documentation of the activities related to the four life cycle phases of a computer based system.

3.2 EXISTING SYSTEM

DRAWBACKS OF EXISTING SYSTEM

3.3 PROPOSED SYSTEM

OBJECTIVES

3.4 FEASIBILITY STUDY

When a new project is proposed, it normally goes through feasibility assessment. Feasibility study is carried out to determine whether the proposed system is possible to develop with available resources and what should be the cost consideration. There are three areas of feasibility study,

- ☐ **Economic Feasibility**
- ☐ **Technical Feasibility**
- ☐ **Operational Feasibility**

The proposed system must be evaluated from a technical view point first, and then considering behavioral and economic feasibility.

3.4.1 ECONOMIC FEASIBILITY

This feasibility study presents tangible and intangible benefits from the project by comparing the development cost and operational cost. This evaluation looks at the financial aspects of the project. The cost to conduct a full system investigation, the cost of hardware and software, the benefits in the form of reduced expenditure are all discussed during the economic feasibility. The technique of cost benefit analysis is often used as a basis for assessing economic feasibility. It determines whether the investments needed to implement the system will be recovered.

The proposed project is economically feasible. Because once the system is put into use in the current market the system provides economical advantage to the firm. Also the firm can afford the cost to implement the project. The proposed project provides tangible and intangible benefits comparing to the existing projects. This system does not need any initial investments and it can improve the quality of service.

3.4.2 TECHNICAL FEASIBILITY

Technical feasibility is related to the software and equipment specified in the design for implementing a new system. It is the study of function, performance and

constraints that may affect the ability to achieve an acceptable system. Technical feasibility includes whether the technology is available on the market for development and its availability. During technical analysis, the analyst evaluates the technical merits of the system, at the same time collecting additional information about performance, reliability, maintainability and productivity. The assessment of technical feasibility must be based on an outline design of system requirements in terms of input, output, files, programs, and procedure. The proposed system meets all the requirement of the Technical feasibility.

Because the implementation of the project need no technological difficult. The web-part of the project can be easily implemented in a user friendly IDE, that is in Android Studio. The proposed project is developed in a 3-tier architecture to provide more security. The system can be easily developed by using the current technologies.

The android application is developed in the most popular language open-source programming language python. The web application is developed in the most popular opensource programming language Python. So the development of the project does not meet any type of the technical difficult. Hence we can be say that the proposed system is technically feasible.

The proposed project is operationally feasible. Because the proposed website is easily operated by the publishers and the subscribers. The proposed web site is operated by the admin and the publishers. The publisher who have basic internet knowledge, he can use the web site for the registration and then use the website for their intended applications. For operating the web content, the publishers need only the basic internet knowledge and an internet connection.

4. SYSTEM SPECIFICATION

4.1 SYSTEM REQUIREMENTS

System requirements are the configuration that a system must have in order for a hardware or software application to run smoothly and efficiently. Failure to meet these requirements can result in installation problems or performance problems. The former may prevent a device or application from getting installed, whereas the latter may cause a product to malfunction or perform below expectation or even to hang or crash. System requirements are also known as minimum system requirements.

There are two types of requirements:

- **Hardware requirements**
- **Software requirement**

4.1.1 HARDWARE REQUIREMENTS

The selection of hardware is very important in the existence and proper working of any software. Then selection hardware, the size and capacity requirements are also important.

- Processor : Intel Pentium Core i3 and above
- Primary Memory : 4GB RAM and above
- Storage : 320 GB hard disk and above
- Display : VGA Colour Monitor
- Key Board : Windows compatible
- Mouse : Windows compatible

4.1.2 SOFTWARE REQUIREMENTS

Software system requirements, in addition to the above mentioned requirements, may also specify additional software dependencies (e.g., libraries, driver version, framework version). For the proposed system to work properly, it is

necessary that following software are installed and running on the server/client.

- Operating System : Windows 8 or above
- Front End : HTML, CSS, Javascript,Bootstrap
- Back End : Python (django)
- Data Base : Mysql
- IDE : Pycharm
- Browser : Internet Explorer/ Mozilla Firefox

5. SOFTWARE DESCRIPTION

5.1 FRONT END

HTML

Hyper Text Markup Language(HTML)is the standard markup language for documents designed to be displayed in a web browser. HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes and other items.

CSS

CSS stands for cascading style sheets.CSS describes how HTML elements are to be displayed on screen, paper, or in other media.CSS saves a lot of work. it can control the layout of multiple web pages all at once. External style sheets are stored in CSS files.

BOOTSTRAP

Bootstrap is a popular open-source front-end framework for developing responsive and mobile-first websites. Originally created by Twitter, it offers a collection of HTML, CSS, and JavaScript tools and components, such as navigation bars, buttons, forms, and carousels, that simplify the process of web development. Bootstrap's grid system allows for the easy creation of complex layouts that adapt seamlessly to different screen sizes. It also provides built-in themes and customization options, enabling developers to create visually appealing and consistent user interfaces quickly.

JAVASCRIPT

JavaScript is a versatile, high-level programming language primarily used for web development. It enables dynamic content, allowing developers to create interactive websites and applications. JavaScript runs on the client side, meaning it executes directly in the user's web browser, which helps reduce server load and enhance user experience. Its capabilities extend beyond the browser, as it can be used on servers through environments like Node.js.

5.2 BACK END

PYTHON

Python is a general-purpose language. it has wide range of applications from Web development(like:Django and Bottle),scientific and mathematical computing (Orange, SymPy, NumPy) to desktop graphical user Interfaces (Pygame,Panda3D).The syntax of the language is clean and length of the code is relatively short. It's fun to work in Python because it allows you to think about the problem rather than focusing on the syntax. Python 3.5 is the last updated version and was on September13,2015.

Features Of Python Programming

- ☐ A simple language which is easier to learn
- ☐ Free and open-source
- ☐ Portability
- ☐ Extensible and Embedded
- ☐ A high-level, interpreted language
- ☐ Object-oriented

DJANGO (FRAMEWORK)

Django is a high-level, open-source web framework written in Python that promotes rapid development and clean, pragmatic design. It follows the model-template-views (MTV) architectural pattern and includes numerous built-in features such as an ORM (Object-Relational Mapping), authentication, and an admin interface, which streamline the development process. Django emphasizes reusability, modularity, and the principle of "don't repeat yourself" (DRY), allowing developers to create robust and scalable web applications quickly and efficiently. With its comprehensive documentation and vibrant community, Django is a popular choice for both beginners and experienced developers.

5.3 DATABASE

MySQL

MySQL, the most popular Open Source SQL database management system, is developed, distributed, and supported by Oracle Corporation. The SQL part of “MySQL” stands for “Structured Query Language”. SQL is the most common standardized language used to access databases. Depending on your programming environment, you might enter SQL directly, embed SQL statements into code written in another language, or use a language-specific API that hides the SQL syntax. SQL is defined by the ANSI/ISO SQL Standard. The SQL standard has been evolving since 1986 and several versions exist.

Features of MySQL

- ☐ MySQL is a database management system.
- ☐ MySQL databases are relational.
- ☐ MySQL software is Open Source.
- ☐ The MySQL Database Server is very fast, reliable, scalable, and easy to use

SQL Commands

SQL commands are instructions, coded into SQL statements, which are used to communicate with the database to perform specific tasks, work, functions and queries with data. SQL commands can be used not only for searching the database but also to perform various other functions like, creating tables, adding values to tables, or modifying data, dropping the table, setting permissions for users. SQL commands are grouped into four major categories depending on their functionality. These SQL commands are used for managing changes affecting the data. These commands are COMMIT, ROLLBACK, and SAVEPOINT.

Data Definition Language (DDL)

These SQL commands are used for creating, modifying, and dropping the structure of database objects. The commands are CREATE, ALTER, DROP, RENAME, and TRUNCATE.

Data Manipulation Language (DML)

These SQL commands are used for storing, retrieving, modifying, and deleting data. These Data Manipulation Language commands are: SELECT, INSERT, UPDATE, and DELETE.

Transaction Control Language (TCL)

These SQL commands are used for managing changes affecting the data. These commands are COMMIT, ROLLBACK, and SAVEPOINT.

Data Control Language (DCL)

These SQL commands are used for providing security to database objects. These commands are GRANT and REVOKE.

6. HARDWARE DESCRIPTION

INPUT DEVICE	:	MOUSE, KEYBOARD(For Web application)
OUTPUT DEVICE	:	MONITOR(For Web application
MEMORY	:	4 GB RAM(MINIMUM)(For Web application)
PROCESSOR	:	INTEL PROCESSOR(For Web application)
WEB BROWSER	:	Internet Explorer/Google Chrome/Firefox (For Web application)

7. PROJECT DESCRIPTION

7.1 PROJECT DESCRIPTION

☐ **Challenges:**

☐ **Applications:**

7.1.a ALGORITHMS

7.2 MODULE DESCRIPTION

MODULES

MODULES AND FUNCTIONS

8. SYSTEM DESIGN

8.1 System Design

Design is the phase that indicates the final system. It is the solution, the translation of requirements into ways of meeting them. In this phase the following elements were designed namely, data flow, data stores, processes, procedures. Firstly the logical design was done where the output, inputs and databases and procedures were formulated in a manner that meets the project requirements. After logical design physical construction of the system is done.

After analyzing the various functions involved in the system the database and tables are designed. Care is taken for the field name to be in self-explanatory form. Unnecessary fields are avoided so as not to affect the storage of the system. Care is taken to design the input screen in the most user friendly way so as to help even the novice users make entries approximately in the right place. All input screens in the system are user friendly. The sizes of all the screens are standardized. The importance of the software design can be stated with a single word quality. Design is a place where quality is fostered in software development. Design is the only way where requirements are actually translated into a finished software product or system. In this project tables are create in MySQL. The first step in the design phase is to design the database and then input and output within predefined guidelines. Simple designs are easily understood, easily built, and easily tested. Simplicity is the most important criteria of a design. Other design criteria include the following

Documentation: A good design always comes with a set of well written documents.

Testability: In a good design every requirement is testable. A design that cannot be easily tested against its requirements is not an acceptable design.

Structure: A good design presents hierarchical structure that makes logical use of control policies among components.

Modularity: A good design is modular and exhibits the properties of high cohesiveness and low coupling

Discreteness: A good design separates data procedures and timing consideration to the extent possible.

Representation: A good design should be easily communicated to all interested parties through appropriate abstraction and representation.

Reusability: A good design should be repeatable and reusable, the major step in design is the preparation of input and the design of output reports in a form which is acceptable to the user.

System Design:

At the First, level the focus is on deciding which modules are needed for the system, the Specification of these modules, how the modules should be interconnected. This is what we call the top level design or system design. The system design describes the characteristics of the system. It has a major impact on the testability and modifiability of a system and also impacts its efficiency.

Detailed Design:

In the second level, the internal design of the modules or how the specifications of the module can be decided upon. The design is called detailed design. Detailed design essentially expands the system design to contain a more detailed description of the processing logic and the data structures such that the design is sufficiently complete for coding.

Modularization

For the development of “Student icon modular approach is used. Modularization is the process of partitioning the system into modular units of limited size to simplify maintenance when necessary. Each and every module is selected according to the function. Modularization makes system development simpler. It is an attempt to minimize the complexity of the problem. Modular design increases the flexibility of the system. The system is partitioned into different segments. The program code is also divided into different subroutines according to the options.

8.2 DATA FLOW DIAGRAM

Demacro, Gane and Sarron introduces Data Flow Diagram. It is an important tool used during system analysis. A DFD model is a system using external entity from which data Flows into a process, while transforms the data and create the output. Data Flows through the other process or external entities or Files. Data in Files may also flow to process as inputs. A DFD shows the usual Flow of data through a system. It views a system as a function that can transform the inputs into desired outputs. The DFD aims to capture the transformations that take place within a system to the input data so that eventually the output data is produced.

The main merit of the DFD is that it can provide an overview of what data the system should process, what transformation of data are done and where the results Flow. So the DFD shows a movement of data through the different transformations or process in the system. Unlike detailed Flow chart, Data Flow Diagrams do not supply detailed description of the modules but graphically describes a system's data and how the data interact with the system.

The purpose of data Flow diagram is to provide a semantic bridge between users and system developers. The diagrams are graphical, eliminating thousands of words, logical representations, modeling what system does, hierarchical, showing systems at any level of details, and jargon less, allowing user understanding and reviewing. The goal of data flow diagram is to have a commonly understood model of a system.

Types of DFD

Data Flow Diagrams are either Logical or Physical.

- **Logical DFD** - This type of DFD concentrates on the system process and flow of data in the system. For example in a Banking software system, how data is moved between different entities.

- **Physical DFD** - This type of DFD shows how the data flow is actually implemented in the system. It is more specific and close to the implementation.

Levels in DFD

In Software engineering DFD (data flow diagram) can be drawn to represent the system of different levels of abstraction. Higher level DFDs are partitioned into low level showing more information and functional elements. Levels in DFD are numbered 0, 1, 2 or beyond. Here, we will see mainly 3 levels in data flow diagram, which are: 0-level DFD, 1-level DFD, and 2-level DFD.

0th -Level DFD:

It is also known as context diagram. It's designed to be an abstraction view, showing the system as a single process with its relationship to external entities. It represents the entire system as single bubble with input and output data indicated by incoming/outgoing arrows.

1st-Level DFD:

In 1-level DFD, context diagram is decomposed into multiple bubbles/processes. In this level we highlight the main functions of the system and breakdown the high level process of 0-level DFD into sub processes.

2nd-Level DFD:

2-level DFD goes one step deeper into parts of 1-level DFD. It can be used to plan or record the specific/necessary detail about the system's functioning

DFD Components

To construct a Data Flow Diagram, we use

- Arrow
- Circles
- Open End Box
- Squares

An arrow identifies the data flow in motion. It is a pipeline through which

information is flown like the rectangle in the flow chart. A circle stands for processthat converts data into information. An open-end box represents a data store, data atrest or a temporary repository of data. A square defines a source or destination of system data.

The graphical representation of the system makes it good communication tool between the user and the analyst. By analyzing the Data Flow diagrams, one can easily track the movement of data and the changes incurred on it during the process. A data flow diagram is a graphical technique that depicts information flow and transforms that are applied as data move from input to output.

The DFD is used to represent increasing information flow and functional details.

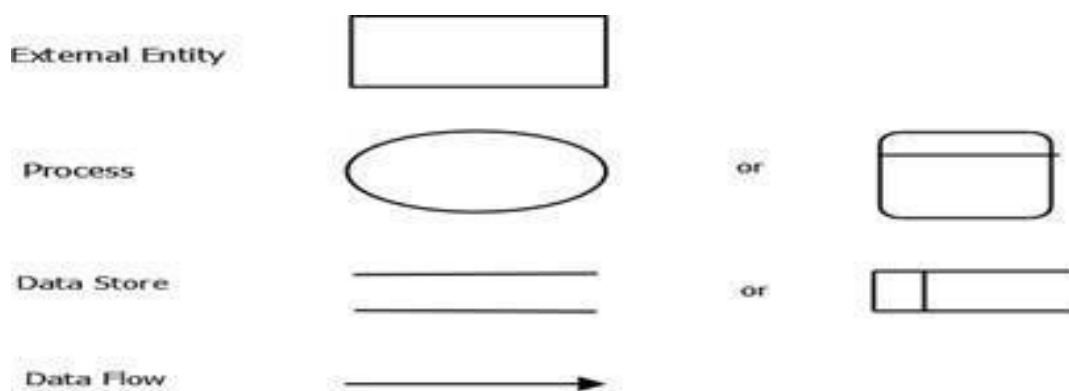


FIG 2:- Source, destination, storage and flow of data

- ❖ **Process:** Specifies the transformation that is applied to the input to produce output i.e. a process transforms incoming data flow into outgoing data flow.
- ❖ **Data Store:** Data stores are repositories of data in the system. They are sometimes also referred to as files.
- ❖ **External Entity:** External entities are objects outside the system, with which the system communicates. External entities are sources and destinations of the system's inputs and outputs. A common example of external entity is person or a group of persons or group of persons.
- ❖ **Data Flow:** This represents the flow of data between two processes or between a process and an external entity or between a process and data store.

Structured analysis is a set of technique and graphical tool that the analyst to develop a new

kind of system specification that are easily understandable to the user. DFD's show the major decomposition of the system functions and their interfaces. The DFD is graphic and presents a picture of what is being specified and is conceptually easy to understand presentation of the application. One important feature of DFD's is that it is logical rather than physical. The elements of the system do not depend on vendor or hardware. They specify in precise, concise manner the working of the systems and how it hangs together.

DFD DESIGNS

LEVEL0

8.3 ER DIAGRAM

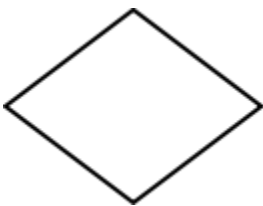
An entity relationship diagram (ERD) shows the relationships of entity sets stored in a database. An entity set is a group of similar entities and these entities can have attributes. In terms of DBMS, an entity is a table or attribute of a table in database, so by showing relationship among tables and their attributes, ER diagram shows the complete logical structure of a database.. ER diagrams are used to sketch out the design of a database.

Basic ER Diagram Symbols and Notations Entity



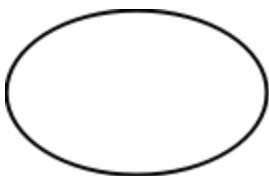
Entity is the thing which we want to store information. An entity represents an objectsdefined within the information system about which you want to store information.

Relationship



A relationship is named connection or association between entities used to relate two ormore entities with some common attributes of meaningful interaction between the object.

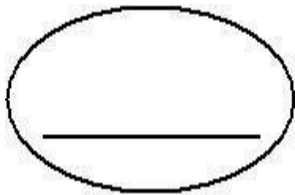
Attributes



Attributes are the properties of the entities and relationship. Descriptor of the entity. Attributes are elementary pieces of information attached to an entity.entity is the thing

which we want to store information. An entity represents an object defined within the information system about which you want to store information.

Key Attributes



The Key attribute is used to denote the property that uniquely identifies an entity (the EntityKey), and which is mapped to the Primary Key field in a database.

DIAGRAM

8.4 DATABASE DESIGN

Database design is one of the most important parts of the system design phase. In a database environment common data are available and are used by several users. Instead of each program managing its own data, authorized users share data across application with the database software managing the data as an entity. The primary objective of a database design is fast response time to enquiries, more information at low cost, control of redundancy, clarity and ease of use, data and program independence, accuracy and integrity of the system, fast recovery and availability of powerful end-user languages. The theme behind a database is to handle information as an integrated whole thus the main objective is to make information as access easy, quick, inexpensive and flexible for the users.

Database design mainly aims at handling large volumes of information, involving the definitions for the structure of storage and provisions for the manipulation of safety of information despite of system crashes due to unauthorized access. Some conditions are satisfied in database design stage.

- Control redundancy
- Ease of use
- Data independency
- Accuracy and integrity
- Recovery from failure Security& privacy
- Performance

The design should be done in a way the information stored in the database can retrieved quickly whenever necessary

8.5 TABLE DESIGN

Table 1: Login Table

Field Name	Data Type	Constraints	Description
Id	Integer (11)	Primary key	User id
Username	Varchar (100)	Not Null	User name
Password	Varchar(100)	Not Null	Password
Type	Varchar(100)	Not Null	Login type

9. SOFTWARE TESTING

Testing is the process of checking whether the developed system is working according to the original objectives and requirements. Software testing is essential for correcting errors. Otherwise the project is not said to be complete. The system should be tested experimentally with test data so as to ensure that the system works according to the required specification. When the system is found working, test it with actual data and check performance. Software testing is a set of processes aimed at investigating, evaluating and ascertaining the completeness and quality of computer software. Software testing ensures the compliance of a software product in relation with regulatory, business, technical, functional and user requirements. Software testing is also known as application testing.

Software testing is primarily a broad process that is composed of several interlinked processes. The primary objective of software testing is to measure software health along with its completeness in terms of core requirements. Software testing involves examining and checking software through different testing processes. The objectives of these processes can include:

- ❑ Verifying software completeness in regards to functional/business requirements
- ❑ Identifying technical bugs/errors and ensuring the software is error-free
- ❑ Assessing usability, performance, security, localization, compatibility and installation

Assessing usability, performance, security, localization, compatibility and installation The tested software must pass each of the tests in-order to be considered complete or fit for use. Some of the different types of software testing methods include white box testing, black box testing and grey box testing. Moreover, the software can be tested as a whole, in components/units or within a live system.

Need For Testing

Testing was essential for the following reasons:

- Existence of program defects or inadequacies
- The software behaviour as intended by its designer
- Conformance with requirement specification/user needs.
- Assesses the operational reliability of the system.
- Reflect the frequency of actual user inputs.
- Find the fault, which caused the output anomaly.
- Checks for detect flows and deficiencies in the requirements.
- Check whether the software is operationally useful.
- Exercise program using data like the real data processed by the program..

Software testing passes through different levels or phases:

- **Unit Testing**
- **Integration Testing**
- **System Testing**
- **Acceptance Testing**
- **White Box Testing**
- **Black Box Testing**

9.1 UNIT TESTING

During this first round of testing, the program is submitted to assessments that focus on specific units or components of the software to determine whether each one is fully functional. The main aim of this endeavor is to determine whether the application functions as designed. In this phase, a unit can refer to a function, individual program or even a procedure, and a White- box Testing method is usually used to get the job done. One of the biggest benefits of this testing phase is that it can be run every time a piece of code is changed, allowing issues to be resolved as quickly as possible. It's quite common for software developers to perform unit tests before delivering software to testers for formal testing.

9.2 INTEGRATION TESTING

Integration testing allows individuals the opportunity to combine all of the units within a program and test them as a group. This testing level is designed to find interface defects between the modules/functions. This is particularly beneficial because it determines how efficiently the units are running together. Keep in mind that no matter how efficiently each unit is running, if they aren't properly integrated, it will affect the functionality of the software program. In order to run these types of tests, individuals can make use of various testing methods, but the specific method that will be used to get the job done will depend greatly on the way in which the units are defined.

9.3 SYSTEM TESTING

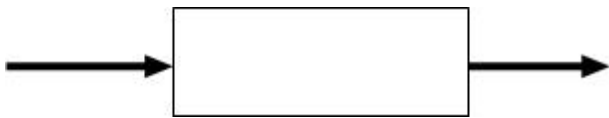
System testing is the first level in which the complete application is tested as a whole. The goal at this level is to evaluate whether the system has complied with all of the outlined requirements and to see that it meets Quality Standards. System testing is undertaken by independent testers who haven't played a role in developing the program. This testing is performed in an environment that closely mirrors production. System Testing is very important because it verifies that the application meets the technical, functional, and business requirements that were set by the customer.

9.4 ACCEPTANCE TESTING

The final level, Acceptance testing (or User Acceptance Testing), is conducted to determine whether the system is ready for release. During the Software development life cycle, requirements changes can sometimes be misinterpreted in a fashion that does not meet the intended needs of the users. During this final phase, the user will test the system to find out whether the application meets their business' needs. Once this process has been completed and the software has passed, the program will then be delivered to production.

9.5 WHITE BOX TESTING

White box testing (also known as Clear Box Testing, Open Box Testing, GlassBox Testing, Transparent Box Testing, Code-Based Testing or Structural Testing) is a software testing method in which the internal structure/design/implementation of the item being tested is known to the tester. The tester chooses inputs to exercise paths through the code and determines the appropriate outputs. Programming know-how and the implementation knowledge is essential. White box testing is testing beyond the user interface and into the nitty-gritty of a system.



This method is named so because the software program, in the eyes of the tester, is like a white/transparent box; inside which one clearly sees.

White Box Testing method is applicable to the following levels of software testing:

- Unit Testing: For testing paths within a unit.
- Integration Testing: For testing paths between units.
- System Testing: For testing paths between subsystems.

However, it is mainly applied to Unit Testing.

9.6 BLACK BOX TESTING

Black Box Testing, also known as Behavioral Testing, is a software testing method in which the internal structure/design/implementation of the item being tested is not known to the tester. These tests can be functional or non-functional, though usually functional.



This method is named so because the software program, in the eyes of the tester, is like a black box; inside which one cannot see. This method attempts to find errors in the following categories:

- ☐ Incorrect or missing functions
- ☐ Interface errors
- ☐ Errors in data structures or external database access
- ☐ Behavior or performance errors
- ☐ Initialization and termination errors

10. SYSTEM IMPLEMENTATION

10.1 SYSTEM IMPLEMENTATION

Implementation is the stage of the project where the theoretical design is turned into a working system. At this stage the main work load, the greatest upheaval and the major impact on the existing system shifts to the user department. If the implementation is not carefully planned and controlled, it can cause confusion.

Implementation includes all those activities that take place to convert from the old system to the new one. The new system may be totally new, replacing an existing manual or automated system or it may be a major modification to an existing system. Proper implementation is essential to provide a reliable system to meet the organization requirements. Successful implementation may not guarantee improvement in the organization using the new system, but improper installation will prevent it.

The process of putting the developed system in actual use is called system implementation. This includes all those activities that take place to convert from the old system to the new system. The system can be implemented only after thorough testing is done and if it is found to be working according to the specifications. The system personnel check the feasibility of the system.

The most crucial stage is achieving a new successful system and giving confidence on the new system for the user that it will work efficiently and effectively. It involves careful planning, investigation of the current system and its constraints on implementation, design of methods to achieve the changeover. The more complex the system being implemented, the more involved will be the system analysis and the design effort required just for implementation. The system implementation has three main aspects. They are education and training, system testing and changeover.

TRAINING

Training is done in two ways,

- User training
- System operator training.

User training: In this system give training for entering the details. That is user details attendance details, leave details, mark details etc. and provide training for giving all options. **System operator training:** It is concerned with how to use the details given by the user and how to maintain all the operations of whole system. Hence providing training for entering new user details, approve leave request, publish internal for the user.

CONVERSION

The crucial stage in conversion to a new successful system and giving confidence on the new system for the user that it will work efficiently and effectively. It involves careful planning, investigation of the current system and its constraints on implementation, design of methods to achieve the changeover. The more complex the system being implemented, the more involved will be the system analysis and the design effort required just for implementation.

POST IMPLEMENTATION AND REVIEW

Implementation is the stage of the project where the theoretical design is turned into a working system. At this stage the main work load, the greatest upheaval and the major impact on the existing system shifts to the user department. If the implementation is not carefully planned and controlled, it can cause confusion. The system implementation has three main aspects. They are education and training, system testing and changeover. Post implementation includes all those activities that take place to convert from the old system to the new one. The new system may be totally new, replacing an existing manual or automated system or it may be a major modification to an existing system. Proper implementation is essential to provide a reliable system to meet the organization requirements. Successful implementation may not guarantee improvement in the organization using the new system, but improper installation will prevent it.

The process of putting the feedback on the developed system in actual use is called System Review. This includes all those activities that take place after the conversion from the old system to the new system. The system can be implemented only after thorough testing is done and if it is found to be working according to the specifications. The system personnel check the feasibility of the system. The most

crucial stage is achieving a new successful system and giving confidence on the new system for the user that it will work efficiently and effectively. It involves careful planning. Investigation of the current system and its constraints on implementation, design of methods to achieve the changeover. The more complex the system being implemented, the more involved will be the system analysis and the design effort required just for implementation.

SECURITY MECHANISM

The system security problem can be divided into four related issues: **security, integrity, privacy and confidentiality**. They determine the file structure, data structure and access procedures. System security refers to the technical innovations and procedures applied to the hardware and operating systems to protect against deliberate or accidental damage from a defined threat. In contrast, data security is the protection of data from loss, disclosure, modifications and destruction.

System integrity refers to the proper functioning of programs, appropriate physical security and safety against external threats such as eavesdropping and wiretapping. In comparison, data integrity makes sure that do not differ from original from others and how the organization can be protected against unwelcome, unfair or excessive dissemination of information about it. The term confidentiality is a special status given to sensitive information in a data base to minimize the possible invasion of privacy. It is an attribute of information that characterizes its need for protection.

10.2 SYSTEM MAINTENANCE

Once the software is fully developed and implemented, the users (management, students, teacher and parents) starts to use the software. The number of users increases on each academic year. So after some time the software, which has been installed, needs some modification. If the software needs modification all the steps needed to develop new software has to be executed.

The need has to be studied, the design has to be made and coding has to be done. Then the new module has to be connected to the existing software modules. Once

the software is working perfect also we have to do routine testing and any new bug is found out, immediately it has to be fixed. No software ever developed will be bug free forever. When a new situation arises, the software can create an error, but if it is found out and repaired the software will not be causing more problems. Always maintenance has to be done on the software, for to make the software work perfectly without any errors. Maintenance can be classified as corrective, adaptive or perceptive. Corrective maintenance means repairing processing or performance failures or making changes because of previously uncorrected problems or false assumptions. Adaptive maintenance means changing the program function.

Perceptive maintenance means enhancing the performance or modifying the programs to respond to the user's additional or changing needs. Maintenance covers a wide range of activities, including correcting, coding and design errors, updating documentation and test data and upgrading user support.

Implementation plan

The implementation plan includes a description of all activities that must occur to implement the new system and to put it into operation. The system is developed as a web application which replaces the existing web application. The different modules in the system are distributed to the users of the system. Administration of the system is the key task of the administrator.

11. CONCLUSION AND FUTURE ENHANCEMENT

11.1 CONCLUSION:

11.2. FUTURE ENHANCEMENT:

12. APPENDIX

APPENDIX

1. SERVER CODE

13. SCREENSHORT

WEB PAGE:

14. REFERENCES

Books

Website